SONIC experiment status

Unitree G1 mimic | seed 42 | 512 environments | verified through July 20, 2026

Key correction

The completed 5,000-iteration SonicLite run used a continuous latent bottleneck. It did not use FSQ. The corrected FSQ model has completed only a 500-iteration pilot, so the old long-run advantage cannot be attributed to quantization.

Current run inventory

Original MLP

COMPLETE

5,000 iterations

Standard observation-to-action MLP baseline.

Final reward: 4.45

Continuous SonicLite

COMPLETE

5,000 iterations

Encoder -> continuous 256-D latent -> decoder. No FSQ.

Final reward: 5.52

SonicLite-FSQ

PILOT

500 iterations

Encoder -> FSQ bottleneck -> decoder. Technically validated.

Final reward: -2.62

Research boundary

All results use one seed. They are useful engineering evidence, not statistical proof. Do not claim that SONIC or FSQ is superior until the faithful architecture and multiple seeds are tested.

Fair 500-iteration pilot comparison

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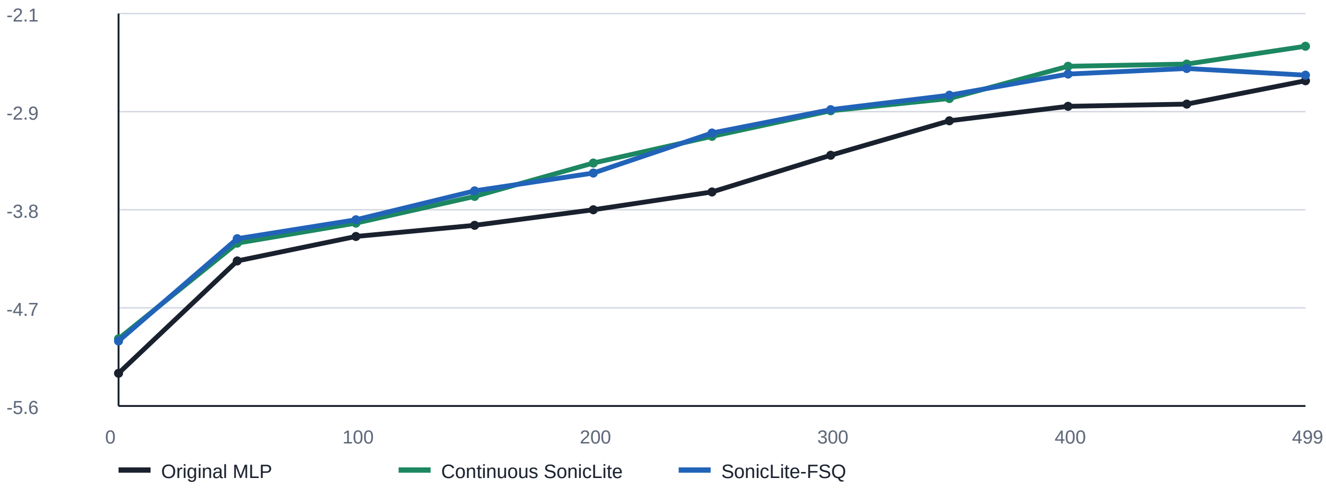
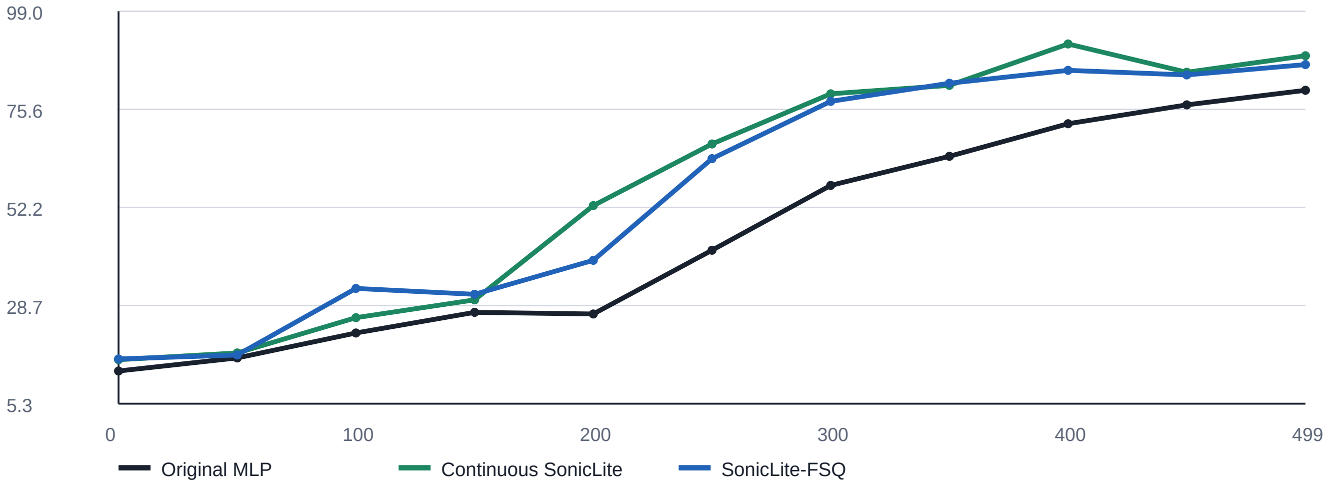
All three runs share the same training budget, seed, environment count, motion data, critic, and PPO settings.

Mean reward during the pilot



X: training iteration | Y: mean episode reward (higher is better)
Source: seed-42 training logs, iterations 0-499, sampled every 50 iterations.

Mean episode length during the pilot



X: training iteration | Y: episode length in simulation steps (higher is better)
Source: seed-42 training logs, iterations 0-499, sampled every 50 iterations.

Pilot metrics and long-run context

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Last-100-iteration pilot averages

Actor	Reward	Ep. length	Action std.	Joint err.	Ang.-vel. err.
Original MLP	-2.7283	77.03	0.6031	0.16177	1.10417
Continuous SonicLite	-2.4077	90.04	0.5858	0.16596	1.06380
SonicLite-FSQ	-2.5156	84.35	0.5836	0.14170	1.05581

Higher reward and episode length are better; lower standard deviation and tracking errors are better.

+0.213
FSQ reward over MLP

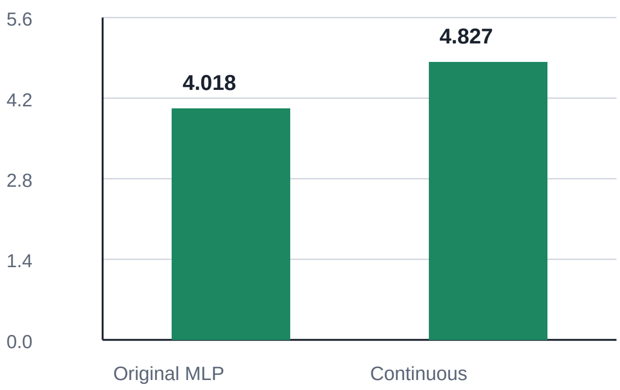
-0.108
FSQ behind continuous

12.4%
FSQ joint error below MLP

What the pilot actually says

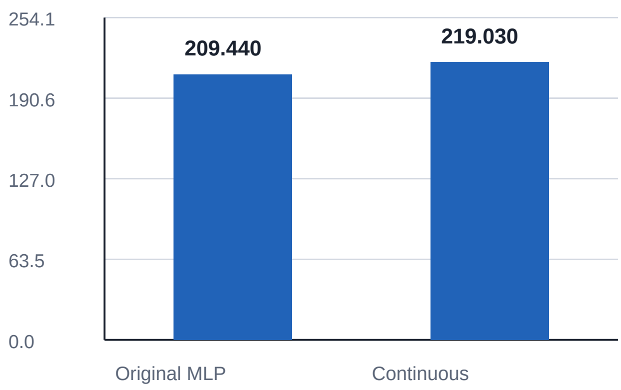
FSQ is functioning and learning. It finished between the two other actors on reward and episode length, while producing the lowest joint-position and angular-velocity tracking errors. At 500 iterations it is promising, but it is not the reward leader.

Final-500 mean reward



5,000-iteration runs; reward units

Final-500 episode length



5,000-iteration runs; simulation steps

FSQ is absent from the long-run charts because no 5,000-iteration FSQ run exists.

Source: iterations 4,500-4,999 of the seed-42 long runs.

Conclusion and next phase

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Verified conclusions

- The prior 5,000-iteration SonicLite advantage belongs to a continuous bottleneck model, not FSQ.
- The corrected FSQ implementation passes the technical pilot and learns under the controlled setup.
- At 500 iterations, continuous SonicLite leads reward and survival; FSQ leads both reported tracking errors.
- Single-seed results do not establish statistical superiority.

Recommended next phase

Move toward the intended SONIC-style architecture before another long run.

1. Represent motion with explicit tokens.
2. Quantize each token directly instead of quantizing one undifferentiated latent vector.
3. Preserve temporal motion structure through the actor.
4. Log codebook usage, occupancy, dead codes, and token stability.
5. Repeat smoke testing and the controlled 500-iteration pilot.
6. Run multiple seeds before making comparative research claims.

Decision gate

Do not start another expensive 5,000-iteration run until the token structure and diagnostics are explicit and the corrected model reproduces stable learning in a controlled pilot.

Report provenance

This PDF is a static export of the Cursor Canvas 'sonic-experiment-status'. All values and interpretations are reproduced from that verified status artifact.